Here is the detailed roadmap for Indian college students to create a low-cost solar-powered lantern with SOS functionality, now including useful references and videos for guidance:

Detailed Roadmap with Resources

1. Research & Planning

- Study simple solar lamp and LED lighting circuits.
- Define core specs: 3 5V solar panel, rechargeable battery, 1W LED, SOS flashing with NE555 timer.
- Sketch circuit and housing plan.

2. Component Sourcing & Budget

Component	Details	Cost INR
Solar Panel	3V, 1.5 3W polycrystalline	100 120
Battery	NiMH AA / 14500 Li-ion, 2000mAh	50 70
White LED	1W	15 20
Switches	SPST or Push-button	10
NE555 Timer IC	For SOS flashing	10 15
Resistors/Diodes	For basic protection	10
Housing material	Cardboard/plastic bottle	Free/recycled
Wiring & Solder	Basic copper wire	10

3. Circuit Design & Prototyping

- Build charging circuit with solar panel and battery protected by diode.
- Add NE555 timer configured for SOS flash (standard "...---..." pattern) controlled by switch.
- Test on breadboard before final assembly.

4. Lantern Housing & Assembly

- Use recycled plastic bottles or cardboard boxes as casing.
- Line inside with aluminum foil for better light diffusion.
- Mount solar panel for optimal sunlight exposure.
- Secure wiring and components safely.

5. Testing & Optimization

- Charge lantern for 6 8 hours in sunlight.
- Test brightness, runtime, and SOS flash visibility.
- Adjust resistor values or battery capacity for optimal performance.

6. Documentation & Sharing

- Record schematic diagrams and parts list.
- Document build steps with photos or videos.
- Share project with peers or online communities.

Useful Resources and Video References

- YouTube: "How to make Automatic ON/OFF Solar Street Lamp Circuit"
  - Explains solar charging circuit basics using transistor and diode components.
  - Video link: https://www.youtube.com/watch?v=bJmbISo3jR0 [1]
- Instructables: "High Powered LED Solar Lighting System"
  - Step-by-step guide for wiring solar panels, LEDs, and basic circuitry.
  - Link: https://www.instructables.com/High-Powered-LED_Solar-Lighting-System/ [2]
- Circuits DIY "Automatic Solar Rechargeable Light Circuit"
  - Tutorial on building automatic solar light circuit with transistors and LEDs.
  - Link: https://www.circuits-diy.com/automatic-solar-rechargeable-light-circuit-diy/ [3]
- YouTube: "Mini Solar Lamp Teardown and Circuit Explanation"
  - Detailed circuit explanation of simple solar lamps.
  - Link: https://www.youtube.com/watch?v=WtXu5Hrx9qo [4]
- Instructables: "Simple Solar Circuits"
  - Multiple solar light circuit options explained with schematics.
  - Link: https://www.instructables.com/Simple-Solar-Circuits/ [5]

These resources offer practical tutorials and circuit diagrams tailored for low-cost solar lighting projects suitable for Indian college students to build and learn hands-on.

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https://www.youtube.com/watch?v=bJmbISo3jR0

https://www.instructables.com/High-Powered-LED_Solar-Lighting-System/

https://www.circuits-diy.com/automatic-solar-rechargeable-light-circuit-diy/

https://www.youtube.com/watch?v=WtXu5Hrx9qo

https://www.instructables.com/Simple-Solar-Circuits/

https://in.pinterest.com/pin/92042386128291809/

https://www.youtube.com/watch?v=1ySoMGwNCGg

https://www.electronicsforu.com/electronics-projects/hardware-diy/solar-powered-home-lighting-system

https://www.youtube.com/watch?v=cX_TtZ1ZjjU

https://rdso.indianrailways.gov.in/works/uploads/File/Handbook on Solar Light System.pdf